LAB # 10 : Sampling Distribution

Introduction

This lab is concerned with the sampling distribution of statistics. The probability distribution of a statistic (sample mean, sample variance etc) is called a sampling distribution.

Sampling distribution of mean

Suppose that a random sample of n observations is taken from a normal population with mean μ and variance σ^2. We should view the sampling distributions of $\overline{X}$ and S^2 as the mechanisms from which we will be able to make inferences on the parameters μ and σ^2. The sampling distribution of $\overline{X}$ with sample size n is the distribution that results when an experiment is conducted over and over (always with sample size n) and the many values of $\overline{X}$ result. This sampling distribution, then, describes the variability of sample averages around the population mean μ.
Each observation $X_i,\ i = 1, 2, \cdots, n$, of the random sample will then have the same normal distribution as the population being sampled. Hence, we conclude that

$$\overline{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$$

has a normal distribution with mean

$$\mu_{\overline{X}} = E(\overline{X}) = \mu$$

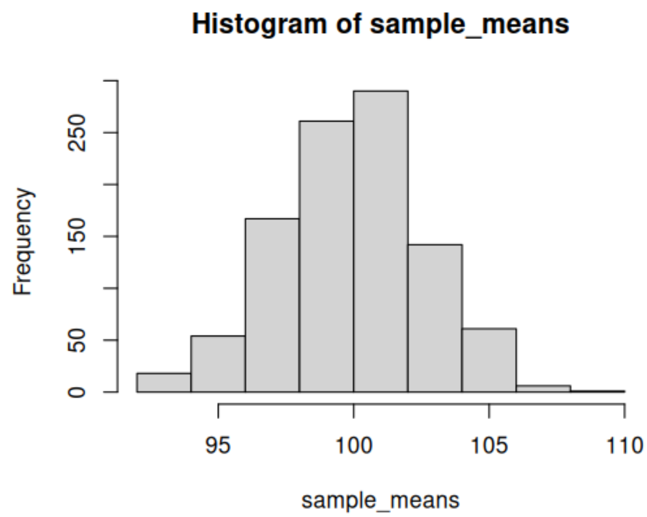
and variance

$$\sigma^2_{\overline{X}} = Var(\overline{X}) = \frac{\sigma^2}{n}.$$

If we are sampling from a population with unknown distribution, either finite or infinite, the sampling distribution of $\overline{X}$ will still be approximately normal with mean μ and variance σ^2/n, provided that the sample size is large. This result is called the Central Limit Theorem.
Example 1: Consider a population with a normal distribution having mean $\mu = 50$ and standard deviation $\sigma = 8$. We take repeated samples of size $n = 30$ and compute the sample means. Also plot the results.

```
set.seed(123)   # for reproducibility
n<- 30
num_samples <- 500
mu<- 50
sigma<- 8
sample_means <- replicate(num_samples,mean(rnorm(n,mean< mu,sd=sigma)))
mean(sample_means)
var(sample_means)
hist(sample_means)
# output
> mean(sample_means)
```

```
[1] 50.00914
> var(sample_means)
[1] 1.958412
> hist(sample_means)
```



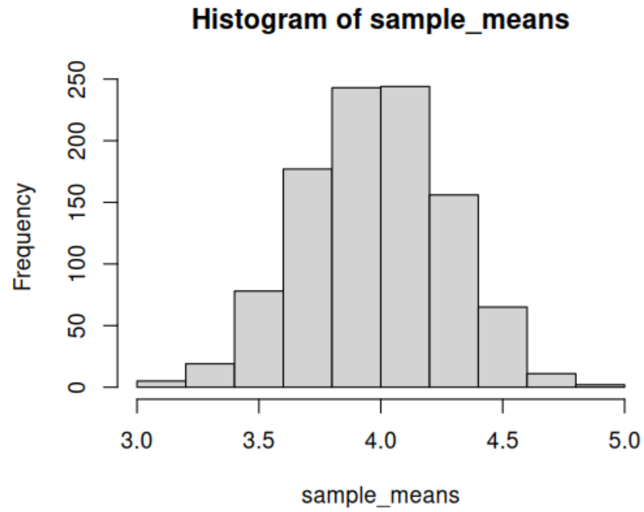
Conclusion: As expected, the sample mean 50.00914 is almost equals to $\mu = 50$ and sample variance is 1.958412 which is close to $\sigma^2/n = 64/30 = 2.13333$.

Example 2: Simulate the sampling distribution of the mean for Binomial distribution with parameters $n = 10$, $p = 0.4$ and plot the results.

```
set.seed(0)
n<- 30
num_samples <- 1000
m<- 10
p <- 0.4
sample_means <- replicate(num_samples,mean(rbinom(n,m,p)))
mean(sample_means)
var(sample_means)
hist(sample_means)
# output
> mean(sample_means)
[1] 3.9939
> var(sample_means)
[1] 0.08729564
> hist(sample_means)
```

Conclusion: As expected, the sample mean almost 4 and sample variance is 0.08729564 which is close to $\sigma^2/n = mpq/30 = 0.08$.

Example 3: Let random variable X follows Gamma distribution with shape parameter 2 and rate parameter 1. Simulate and compare the sampling distribution for $n = 10$ and $n = 100$. Plot the results.

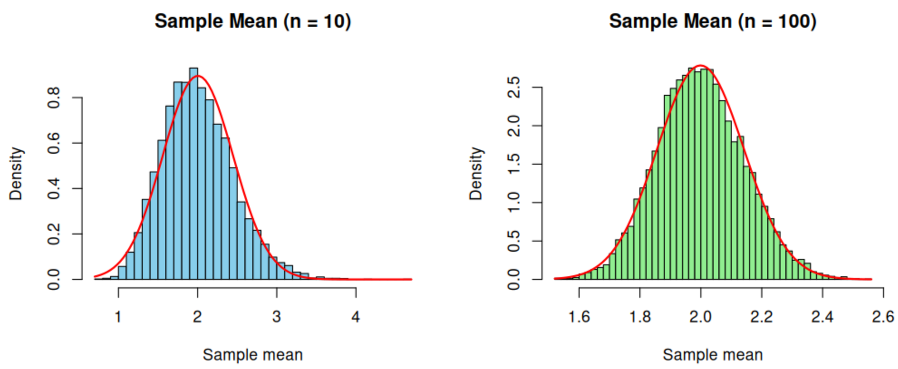



```
set.seed(123) # for reproducibility
n_simulations <- 10000
# Simulation function
simulate_means <- function(n, shape = 2, rate = 1, n_sim = 10000) {
 replicate(n_sim, mean(rgamma(n, shape = shape, rate = rate)))
            }
# Simulate for n = 10 and n = 100
means_n10 <- simulate_means(n = 10)
means_n100 <- simulate_means(n = 100)
# Plot the results
 par(mfrow = c(1,2)) # side-by-side plots
 hist(means_n10, breaks = 40, col = "skyblue", main = "Sample Mean (n
     =10)", xlab = "Sample mean", probability = TRUE)
      curve(dnorm(x, mean = mean(means_n10), sd = sd(means_n10)), col = "red",
      add=TRUE, lwd = 2)
 hist(means_n100,breaks = 40, col = "lightgreen",main ="Sample Mean (n = 100)"
     xlab = "Sample mean", probability = TRUE)
     curve(dnorm(x, mean = mean(means_n100), sd = sd(means_n100)), col = "red"
     add=TRUE, lwd = 2)
  par(mfrow = c(1,1)) # reset plot layout
```

Computing Probabilities for the Variance

Take a sample of 18 having a population mean of 30 and a population variance of 90. What is the probability that S^2 will be less than 160?

```
n <- 18
pop.var <- 90
value <- 160
```




```
pchisq((n - 1) * value/pop.var, n - 1)
[1] 0.9752137
```

Now consider the fruit company problem with weight of apple sauce in grams having normal distribution with mean 275 and variance 0.0016. Here we want to take a random sample of 9 jars and find the s^2 so that $P(S^2 \leq s^2) = 0.99$. The following R command does this:

```
pop.var = 0.0016
n = 9
prob = 0.99
pop.var*qchisq(prob, n - 1)/(n - 1)
[1] 0.004018047
```

Note that the four functions d, p, q, r are valid for the chi-squared distribution. We illustrate by examples.

```
# dchisq(x, df,...)   df is n-1
dchisq(1:5, 6)
[1] 0.03790817 0.09196986 0.12551072 0.13533528 0.12825781
pchisq(1:5, 6)
[1] 0.01438768 0.08030140 0.19115317 0.32332358 0.45618688
qchisq(c(0.50,0.75), 6)
[1] 5.348121 7.840804
rchisq(10,6)
[1]  1.635207  5.402772  3.776536  7.149582 10.340894  1.402825
[7]  1.927993  6.239156  7.197403  1.364009
```

Exercises:

1. If random variable X has Poisson distribution with $\lambda = 4$, simulate the sampling distribution of the mean for sample size $n = 50$. Compare theoretical and empirical mean and variance.

2. If X follows exponential distribution with $\lambda = 1.5$, simulate sample means and check normality for $n = 10$ and $n = 50$.